

1st order, linear constant coefficient ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = f(x)$$

Solving the 1st order, linear, constant coefficient homogeneous ODE

Recall the auxiliary eqn method used on
1st order, linear, constant coefficient, homogeneous ODE
where we know by hand that the solution for it
is $y = Ae^{\lambda x}$.

Now for 1st order, linear, constant-coefficient homogeneous ODE,
let assume $y = Ae^{\lambda x}$ is a solution.

$$\text{Given } \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

$$y = Ae^{\lambda x}, \quad \frac{dy}{dx} = \lambda Ae^{\lambda x}, \quad \frac{d^2 y}{dx^2} = \lambda^2 Ae^{\lambda x}$$

Substituting these we get,

$$\lambda^2 (Ae^{\lambda x}) - 5\lambda (Ae^{\lambda x}) + 6(Ae^{\lambda x}) = 0$$

$$\lambda^2 - 5\lambda + 6 = 0 \quad [\text{since } Ae^{\lambda x} \neq 0]$$

Auxiliary Eqn

Similar to the 1st order counterpart, we can just immediately

write the auxiliary Eqn by changing

$$\frac{d^2 y}{dx^2} \rightarrow \lambda^2, \quad \frac{dy}{dx} \rightarrow \lambda, \quad y \rightarrow 1$$

Continue solving,

$$(\lambda - 3)(\lambda - 2) = 0$$

$$\lambda = 3 \text{ or } \lambda = 2$$

$$\therefore y = Ae^{3x} \text{ or } y = Be^{2x}$$

Solving 1st order, linear, constant-coefficient, homogeneous ODE

$$\begin{aligned} \frac{dy}{dx} + \lambda y &= 0 \\ \left(\frac{y}{x}\right) \frac{dy}{dx} &= -\lambda \\ \int \frac{1}{y} dy &= -\lambda \int \frac{1}{x} dx \\ \ln|y| &= -\lambda x + c \\ y &= \pm e^{-\lambda x + c} \\ &= (\pm e^c)(e^{-\lambda x}) \\ &= Ae^{-\lambda x} \text{ where } A = \pm e^c \end{aligned}$$

Example 1:

$$5 \frac{dy}{dx} + y = 0 \quad \text{--- } \textcircled{1}$$

Since this is a 1st order, linear, constant-coefficient, homogeneous ODE,
we know for sure $y = Ae^{\lambda x} \Rightarrow \frac{dy}{dx} = \lambda Ae^{\lambda x}$

$$\begin{aligned} \therefore \textcircled{1} \text{ becomes } 5[\lambda Ae^{\lambda x}] + Ae^{\lambda x} &= 0 \\ Ae^{\lambda x} [5\lambda + 1] &= 0 \\ 5\lambda + 1 &= 0 \quad [\text{Auxiliary Eqn}] \\ \lambda &= -\frac{1}{5} \end{aligned}$$

$$\therefore \text{General solution: } y = Ae^{-\frac{1}{5}x}$$

* Note

$$\begin{aligned} \text{Let } \frac{dy}{dx} + ky &= 0, \text{ and we know } y = Ae^{\lambda x} \Rightarrow \frac{dy}{dx} = \lambda Ae^{\lambda x} \\ \therefore \lambda Ae^{\lambda x} + kAe^{\lambda x} &= 0 \\ \lambda + k &= 0 \quad [\text{Auxiliary Eqn}] \end{aligned}$$

$$\therefore \text{We can immediately write the auxiliary Eqn by changing } \frac{dy}{dx} \rightarrow \lambda, \quad y \rightarrow 1$$

2nd order must have
2 arbitrary constants

$\therefore$ General sol. is

$$y = Ae^{3x} + Be^{2x}, \quad A, B \text{ are arbitrary constants}$$

Why is general sol $y = y_1 + y_2$?

$$\text{Let } y_{\text{sum}} = y_1 + y_2$$

$$\frac{d}{dx}(y_{\text{sum}}) = \frac{d}{dx}(y_1 + y_2)$$

$$= \frac{d}{dx}y_1 + \frac{d}{dx}y_2$$

$$\frac{d}{dx}(y_{\text{sum}}) = \frac{d}{dx}\left[\frac{dy_1}{dx} + \frac{dy_2}{dx}\right]$$

$$= \frac{d^2}{dx^2}(y_1) + \frac{d^2}{dx^2}(y_2)$$

$$y_1 \text{ is a soln}$$

$$\frac{d^2 y_1}{dx^2} - 5 \frac{dy_1}{dx} + 6y_1 = 0 \quad \text{--- } \textcircled{1}$$

$$y_2 \text{ is a soln}$$

$$\frac{d^2 y_2}{dx^2} - 5 \frac{dy_2}{dx} + 6y_2 = 0 \quad \text{--- } \textcircled{2}$$

$$\textcircled{1} + \textcircled{2}:$$

$$\frac{d^2}{dx^2}(y_1 + y_2) - 5 \frac{d}{dx}(y_1 + y_2) + 6(y_1 + y_2) = 0$$

$$\frac{d^2}{dx^2}(y_{\text{sum}}) - 5 \frac{d}{dx}(y_{\text{sum}}) + 6(y_{\text{sum}}) = 0$$

There are **three types of solution to this quadratic equation** (two distinct real roots / two equal real roots / two complex roots which are conjugates of each other), and **they each lead to different types of solution for the differential equation**.

<i>Value of discriminant</i>	<i>Type of roots for the auxiliary equation</i>	<i>Complementary function</i>
$b^2 - 4ac > 0$	Real, distinct roots, λ_1 and λ_2	$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
$b^2 - 4ac = 0$	Real, repeated root, λ	$y = (A + Bx)e^{\lambda x}$ <i>[just memorise]</i>
$b^2 - 4ac < 0$	Complex roots: (a) $\alpha \pm \beta i$, with $\alpha \neq 0$ (b) $\pm \beta i$ (i.e. purely imaginary roots)	(a) $y = e^{\alpha x}(P \cos \beta x + Q \sin \beta x)$ (b) $y = P \cos \beta x + Q \sin \beta x$

Example 4 (Auxiliary equation with distinct real roots, $b^2 - 4ac > 0$)

Find the **particular solution** of the differential equation $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 0$ subject to the conditions $y = 0$ and $\frac{dy}{dx} = 1$ when $x = 0$.
 (Ans: $y = \frac{1}{2}e^{-x} - \frac{1}{2}e^{-3x}$)

[Hint: Find the general solution first. Then, use the conditions to find the values of the constants A and B .]

Solution

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 0$$

Auxiliary eqn

$$\lambda^2 + 4\lambda + 3 = 0$$

$$(\lambda + 3)(\lambda + 1) = 0$$

$$\lambda = -3 \text{ or } \lambda = -1$$

Just some terminology

The Complementary func.

$$y = Ae^{-3x} + Be^{-x}$$

A complementary function

A complementary function

General soln

$$y = Ae^{-3x} + Be^{-x} \text{ --- (1)}$$

$$\frac{dy}{dx} = -3Ae^{-3x} - Be^{-x} \text{ --- (2)}$$

When $x = 0, y = 0$,

$$0 = Ae^0 + Be^0 \text{ [using (1)]}$$

$$0 = A + B \text{ --- (3)}$$

When $x = 0, \frac{dy}{dx} = 1$,

$$1 = -3Ae^0 - Be^0 \text{ [using (2)]}$$

$$1 = -3A - B \text{ --- (4)}$$

(3) + (4),

$$1 = -2A$$

$$A = -\frac{1}{2}$$

Sub $A = -\frac{1}{2}$ into (3):

$$0 = -\frac{1}{2} + B$$

$$B = \frac{1}{2}$$

$\therefore$ Particular soln:

$$y = -\frac{1}{2}e^{-3x} + \frac{1}{2}e^{-x}$$

Example 5 (Auxiliary equation with repeated real root, $b^2 - 4ac = 0$)

Find the **particular solution** of the differential equation $\frac{d^2z}{dt^2} + 6\frac{dz}{dt} + 9z = 0$ subject to the conditions $z = 0$ and $\frac{dz}{dt} = 5$ when $t = 0$.
 (Ans: $z = 5te^{-3t}$)

Solution

$$\frac{d^2z}{dt^2} + 6\frac{dz}{dt} + 9z = 0$$

Auxiliary Eqn

$$\lambda^2 + 6\lambda + 9 = 0$$

$$(\lambda + 3)^2 = 0$$

$$\lambda = -3 \text{ [Repeated]}$$

Complementary func.

$$z = Ae^{-3t} + Bte^{-3t}$$

$$= (A + Bt)e^{-3t}$$

If this was not added,

$$z = Ae^{-3t} + Be^{-3t}$$

$$= Ce^{-3t}, \text{ where } C = A + B$$

However, 2nd order must have 2 constants, thus this is invalid.

General soln:

$$z = (A + Bt)e^{-3t} \text{ --- (1)}$$

$$\frac{dz}{dt} = -3(A + Bt)e^{-3t} + Be^{-3t} \text{ --- (2)}$$

When $z = 0, t = 0$,

$$0 = (A + 0)e^0 \text{ [using (1)]}$$

$$A = 0$$

When $t = 0, \frac{dz}{dt} = 5$

$$5 = -3(0 + 0)e^0 + Be^0$$

$$B = 5$$

$\therefore$ Particular soln:

$$z = 5te^{-3t}$$

Example 6 (Auxiliary equation with purely imaginary roots, $\lambda = \pm \beta i$)

- (a) Find the **general solution** of the differential equation $\frac{d^2x}{dt^2} + 4x = 0$. (Ans: $x = A \cos 2t + B \sin 2t$)
- (b) Find the particular solution in the case for which $x = 2$ and $\frac{dx}{dt} = -2$ when $t = 0$. (Ans: $x = 2 \cos 2t - \sin 2t$)

Solution

$$a) \quad \frac{d^2x}{dt^2} + 4x = 0$$

Auxiliary eqn

$$\lambda^2 + 4 = 0$$

$$\lambda^2 = -4$$

$$\lambda = \pm 2i$$

Complementary func

$$x = A \cos(2t) + B \sin(2t)$$

General Soln:

$$x = A \cos(2t) + B \sin(2t) \quad \text{--- ①}$$

$$b) \quad \frac{dx}{dt} = -2A \sin(2t) + 2B \cos(2t) \quad \text{--- ②}$$

when $x = 2, t = 0$

$$2 = A \cos(0) + B \sin(0) \quad [\text{using ①}]$$

$$2 = A(1) + 0$$

$$\boxed{A = 2}$$

When $t = 0, \frac{dx}{dt} = -2$

$$-2 = -2A(0) + 2B(1)$$

$$-2 = 2B$$

$$\boxed{B = -1}$$

 $\therefore$ Particular soln:

$$x = 2 \cos 2t - \sin 2t$$

Example 7 (Auxiliary equation with complex roots, $\lambda = \alpha \pm \beta i$)

Find the **particular solution** of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$ which satisfies the initial conditions $y = 0$ and $\frac{dy}{dx} = 1$ when $x = 0$. (Ans: $y = \frac{1}{2} e^{-x} \sin 2x$)

Solution

Auxiliary eqn

$$\lambda^2 + 2\lambda + 5 = 0$$

$$\lambda = \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2(1)}$$

$$= \frac{-2 \pm \sqrt{-16}}{2}$$

$$= \frac{-2 \pm 4i}{2}$$

$$= -1 \pm 2i$$

Complementary func.

$$y = e^{-x} [A \cos(2x) + B \sin(2x)]$$

General eqn

$$y = e^{-x} [A \cos(2x) + B \sin(2x)] \quad \text{--- ①}$$

$$\frac{dy}{dx} = e^{-x} [-2A \sin(2x) + 2B \cos(2x)] - e^{-x} [A \cos(2x) + B \sin(2x)] \quad \text{--- ②}$$

when $y = 0, x = 0$,

$$0 = e^0 [A(1) + B(0)]$$

$$\boxed{A = 0}$$

when $x = 0, \frac{dy}{dx} = 1$,

$$1 = e^0 [-2A(0) + 2B(1)] - e^0 [A(1) + B(0)]$$

$$1 = 2B - A$$

$$1 = 2B - 0$$

$$\boxed{B = \frac{1}{2}}$$

 $\therefore$ Particular soln:

$$y = e^{-x} \left[\frac{1}{2} \sin(2x) \right]$$

$$= \frac{1}{2} e^{-x} \sin 2x \quad \#$$

Solving 1st Order, Non-homo, linear ODE

$$a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = f(x)$$

Just like solving 1st order, non-homo, linear ODE, the solution is

$$y = y_c + y_p$$

The complementary function Particular Integral

3. Procedure for finding the Particular Integral y_p :

- (1) Make a "guess" as to the form of y_p . We call such a "guess" a *trial function*.
- (2) Adjust the trial function to **eliminate any overlap with y_c** .
- (3) **Plug our trial function back into the original differential equation, and then solve for the unknown coefficients.**

This procedure for finding the particular integral y_p is known as the *Method of Undetermined Coefficients*.

4. The table below is a **guide to what trial function to use** for the differential equation

$$a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = f(x)$$

Right-hand side (i.e. the $f(x)$)	Trial function
Linear function	$Px + Q$
Polynomial of degree n	$P_n x^n + P_{n-1} x^{n-1} + \dots + P_1 x + P_0$ (typically we will just use letters P, Q, R , etc rather than using subscripts on the same letter)
Trigonometric function involving $\cos kx$ and/or $\sin kx$	$P \cos kx + Q \sin kx$
Exponential function involving e^{kx}	$P e^{kx}$
Sum of different functions	Sum of matching functions

You should note that this method finds a simple particular integral, but actually there are infinitely many possible particular integrals, which can be constructed by adding on any term from the complementary function.

Multiplication of exponential func.

(Polynomial of deg n) e^{kx}	$y_p(x) = [P_n x^n + P_{n-1} x^{n-1} + \dots + P_0] e^{kx}$
(Trig involving $\cos mx$ and/or $\sin mx$) (e^{kx})	$y_p(x) = [P \cos(mx) + Q \sin(mx)] e^{kx}$
(Polynomial deg n) (Trigo of $\cos mx$ and/or $\sin mx$)	$y_p(x) = (Ax^n + Bx^{n-1} + \dots + C) \cos(mx) + (Dx^n + Ex^{n-1} + \dots + F) \sin(mx)$

TABLE 4.4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

5. Overlap with y_c :

In some differential equations, the **function on the right-hand side** has the **same form as one of the complementary functions**. y_c & y_p must be linearly independent [can't have same variable expressions]

To deal with such situations, we **multiply the functions in y_p** that cause the overlap **by the independent variable (usually x)** until we have eliminated the overlap. *not individual terms.*

For an n^{th} order differential equation, n is the maximum number of times you would need to multiply by x to eliminate overlap.

[Advice: In order to recognise such overlaps when they arise, you should always **find the complementary function y_c before you find the particular integral y_p** .]

Example 8

Find the **general solution** of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 6x - 2$$

$$(Ans: y = Ae^{3x} + Be^{-x} - 2x + 2)$$

Solution

$$\begin{aligned}\text{Auxiliary Eqn: } \lambda^2 - 2\lambda - 3 &= 0 \\ (\lambda - 3)(\lambda + 1) &= 0 \\ \lambda &= 3 \text{ or } \lambda = -1\end{aligned}$$

$$\text{Complementary func, } y_c = Ae^{3x} + Be^{-x}$$

$$\text{Particular integral, } y_p = Px + Q$$

$$y_p' = P$$

$$y_p'' = 0$$

$$\therefore y_p'' - 2y_p' - 3y_p = 6x - 2$$

$$0 - 2P - 3(Px + Q) = 6x - 2$$

$$(-3P)x + (-2P - 3Q) = 6x - 2$$

By comparison,

$$-3P = 6$$

$$-2P - 3Q = -2$$

$$\boxed{P = -2}$$

$$-2(-2) - 3Q = -2$$

$$\boxed{Q = 2}$$

$$\therefore y_p = -2x + 2$$

General soln

$$\therefore y = y_c + y_p$$

$$= Ae^{3x} + Be^{-x} + (-2x + 2)$$

$$= Ae^{3x} + Be^{-x} - 2x + 2 \quad \#$$

Example 9 (Dealing with overlap with y_c)

The differential equation

$$\frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 16y = 2e^{-4t} + t$$

has initial conditions $y = 1, \frac{dy}{dt} = 2$ when $t = 0$. Find the **particular solution**.

$$(Ans: y = (t^2 + \frac{97}{16}t + \frac{33}{32})e^{-4t} + \frac{1}{16}t - \frac{1}{32})$$

Solution

Auxiliary Eqn

$$\lambda^2 + 8\lambda + 16 = 0$$

$$(\lambda + 4)^2 = 0$$

$$\lambda = -4 \text{ (repeated)}$$

Complementary func.

$$y_c = Ae^{-4t} + Bte^{-4t}$$

$$= (A + Bt)e^{-4t}$$

Particular integral

Since y_p overlaps with y_c ,
we multiply y_p by t until
there is no more overlap

$$y_p = y_{p1} + y_{p2}$$

$$y_p = (Pt^2e^{-4t}) + (Qt + R)$$

$$y_p' = [Pt^2(-4e^{-4t}) + (2Pt)e^{-4t}] + Q$$

$$= [2Pt - 4Pt^2]e^{-4t} + Q$$

$$y_p'' = [(2Pt - 4Pt^2)(-4e^{-4t}) + (2P - 8Pt)(e^{-4t})] + 0$$

$$= (-8Pt + 16Pt^2 + 2P - 8Pt)e^{-4t}$$

$$= (16Pt^2 - 16Pt + 2P)(e^{-4t})$$

$$\therefore y_p'' + 8y_p' + 16y_p = 2e^{-4t} + t$$

$$\Rightarrow [16Pt^2 - 16Pt + 2P]e^{-4t} + [16Pt - 32Pt^2]e^{-4t} + 16Pt^2e^{-4t} + 16Qt + 16R = 2e^{-4t} + t$$

$$\Rightarrow [2Pe^{-4t} + (16Q)t + (8Q + 16R)] = 2e^{-4t} + t + 0$$

By comparison

$$2P = 2 \Rightarrow \boxed{P = 1}$$

$$16Q = 1 \Rightarrow \boxed{Q = \frac{1}{16}}$$

$$8Q + 16R = 0$$

$$\Rightarrow 8(\frac{1}{16}) + 16R = 0$$

$$\Rightarrow \boxed{R = -\frac{1}{32}}$$

 $\therefore$ Particular integral is

$$y_p = t^2e^{-4t} + \frac{1}{16}t - \frac{1}{32}$$

General soln

$$y = y_c + y_p$$

$$= (A + Bt)e^{-4t} + t^2e^{-4t} + \frac{1}{16}t - \frac{1}{32}$$

$$= (A + Bt + t^2)e^{-4t} + \frac{1}{16}t - \frac{1}{32}$$

$$\frac{dy}{dt} = (A + Bt + t^2)(-4e^{-4t}) + (B + 2t)(e^{-4t}) + \frac{1}{16}$$

When $t = 0, y = 1$

$$1 = A - \frac{1}{32}$$

$$A = \frac{33}{32}$$

When $t = 0, \frac{dy}{dt} = 2$

$$2 = \frac{33}{32}(-4) + B + \frac{1}{16}$$

$$B = \frac{97}{16}$$

Particular Soln

$$y = [\frac{33}{32} + \frac{97}{16}t + t^2]e^{-4t} + \frac{1}{16}t - \frac{1}{32}$$

Example 10 (CIE 2014s11/4)Obtain the **general solution** of the differential equation

$$\frac{d^2x}{dt^2} - 6\frac{dx}{dt} + 25x = 195 \sin 2t$$

[6]

$$(Ans: x = e^{3t}(A \cos 4t + B \sin 4t) + 7 \sin 2t + 4 \cos 2t)$$

Solution

Auxiliary Eqn

$$\lambda^2 - 6\lambda + 25 = 0$$

$$\lambda = \frac{6 \pm \sqrt{6^2 - 4(1)(25)}}{2(1)}$$

$$= 3 \pm 4i$$

Complementary func.

$$x_c = e^{3t}(A \cos 4t + B \sin 4t)$$

Particular Integral

$$x_p = P \cos 2t + Q \sin 2t$$

$$x_p' = -2P \sin 2t + 2Q \cos 2t$$

$$x_p'' = -4P \cos 2t - 4Q \sin 2t$$

$$\therefore x_p'' - 6x_p' + 25x_p = 195 \sin 2t$$

$$\Rightarrow (21P - 12Q) \cos 2t + (21Q + 12P) \sin 2t = 0 \cos 2t + 195 \sin 2t$$

By comparison

$$21P - 12Q = 0 \Rightarrow \boxed{Q = \frac{7}{4}P}$$

$$21Q + 12P = 195$$

$$21\left[\frac{7}{4}P\right] + 12P = 195$$

$$\boxed{P = 4}$$

$$\therefore P = 4, Q = 7$$

General soln

$$x = x_c + x_p$$

$$= e^{3t}(A \cos 4t + B \sin 4t) + 7 \sin 2t + 4 \cos 2t$$

Example 3:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = (x+1)e^x$$

Auxiliary Egn

$$\lambda^2 - 3\lambda + 2 = 0$$

$$(\lambda - 2)(\lambda - 1) = 0$$

$$\lambda = 1 \text{ or } \lambda = 2$$

Complementary Function

$$y_c = Ae^x + Be^{2x}$$

Since the variable expression of e^x is in y_c , y_p can't have the same variable expression of e^x . Thus we multiply e^x w x to make the variable expression for y_p , xe^x

Particular Integral

$$y_p = [Px + R][xe^x] \\ = [Px^2 + Rx]e^x$$

$$y_p' = [2Px + R]e^x + [Px^2 + Rx]e^x \\ = [Px^2 + (2P+R)x + R]e^x$$

$$y_p'' = [2Px + (2P+R)]e^x + [Px^2 + (2P+R)x + R]e^x \\ = [Px^2 + (4P+R)x + (2P+2R)]e^x$$

$$\therefore [Px^2 + (4P+R)x + (2P+2R)]e^x \\ - 3[Px^2 + (2P+R)x + R]e^x \\ + 2[Px^2 + Rx]e^x = (x+1)e^x \\ \Rightarrow [-1Px + (2P-R)]e^x = (x+1)e^x$$

By comparison,

$$-1P = 1 \quad 2(-1/2) - R = 1 \\ P = -1/2 \quad R = -1$$

$$\therefore y_p = (-\frac{1}{2}x^2 - 1x)e^x$$

General soln

$$y = y_c + y_p \\ = Ae^x + Be^{2x} + [-\frac{1}{2}x^2 - 2x]e^x \quad \neq$$

5.3: USE OF SUBSTITUTIONS TO TRANSFORM DIFFERENTIAL EQUATIONS

Under this subtopic, we will learn to **use a given substitution to reduce** a differential equation to a **first or second order linear equation with constant coefficients** or to a **first order equation with separable variables**.

Example 11 (CIE 2010w1/11)

It is given that $x \neq 0$ and

$$x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 4xy = 8x^2 + 16.$$

- 1) Show that if $z = xy$ then

$$\frac{d^2 z}{dx^2} + 4z = 8x^2 + 16.$$

[3]

- 2) Find y in terms of x , given that $y = 0$ and $\frac{dy}{dx} = -2$ when $x = \frac{1}{2}\pi$.

[9]

$$\left(\text{Ans: } y = \frac{1}{x} \left[\left(\frac{\pi^2}{2} + 3 \right) \cos 2x + \frac{3\pi}{2} \sin 2x \right] + 2x + \frac{3}{x} \right)$$

Solution

1) $z = xy$ ——— ①

$$\frac{dz}{dx} = x \frac{dy}{dx} + y \text{ ——— ②}$$

$$\begin{aligned} \frac{d^2 z}{dx^2} &= x \left[\frac{d^2 y}{dx^2} \right] + (1) \left[\frac{dy}{dx} \right] + \frac{dz}{dx} \\ &= x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} \end{aligned}$$

$$\therefore x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 4xy = 8x^2 + 16$$

$$\frac{d^2 z}{dx^2} + 4z = 8x^2 + 16 \quad \#$$

2) Auxiliary Eqn

$$\lambda^2 + 4 = 0$$

$$\lambda = \pm \sqrt{-4}$$

$$= \pm 2i$$

Complementary func

$$z_c = A \cos 2x + B \sin 2x$$

Particular Integral

$$z_p = Px^2 + Qx + R$$

$$z_p' = 2Px + Q$$

$$z_p'' = 2P$$

$$\therefore z_p'' + 4z_p = 8x^2 + 16$$

$$\Rightarrow 2P + 4Px^2 + 4Qx + 4R = 8x^2 + 16$$

$$\Rightarrow (4P)x^2 + (4Q)x + (2P + 4R) = 8x^2 + 0x + 16$$

By comparison

$$4P = 8 \Rightarrow \boxed{P = 2}$$

$$4Q = 0 \Rightarrow \boxed{Q = 0}$$

$$2P + 4R = 16$$

$$\Rightarrow P + 2R = 8$$

$$2 + 2R = 8$$

$$\boxed{R = 3}$$

$\therefore$ Particular Integral is

$$z_p = 2x^2 + 3$$

General Soln

$$z = z_c + z_p$$

$$z = A \cos 2x + B \sin 2x + 2x^2 + 3 \text{ ——— ③}$$

$$\frac{dz}{dx} = -2A \sin 2x + 2B \cos 2x + 4x \text{ ——— ④}$$

$$\text{When } x = \frac{\pi}{2}, y = 0, \frac{dy}{dx} = -2$$

$$z = \left(\frac{\pi}{2}\right)(0) = 0 \text{ [using ③]}$$

$$\therefore 0 = A \cos \pi + B \sin \pi + 2\left(\frac{\pi^2}{4}\right) + 3 \text{ [using ③]}$$

$$\boxed{A = \frac{\pi^2}{2} + 3}$$

$$\frac{dz}{dx} = \left(\frac{\pi}{2}\right)(-2) + 0 = -\pi \text{ [using ④]}$$

$$\therefore -\pi = -2A \sin \pi + 2B \cos \pi + 4\left(\frac{\pi}{2}\right) \text{ [using ④]}$$

$$\boxed{B = \frac{3\pi}{2}}$$

Particular soln

$$z = \left(\frac{\pi^2}{2} + 3\right) \cos 2x + \frac{3\pi}{2} \sin 2x + 2x^2 + 3$$

$$y = \frac{1}{x} \left[\left(\frac{\pi^2}{2} + 3\right) \cos 2x + \frac{3\pi}{2} \sin 2x + 2x^2 + 3 \right] \quad \#$$

Example 12 (CIE 2011w11/11O)

Given that

$$x^2 \frac{d^2 y}{dx^2} + 4x(1+x) \frac{dy}{dx} + 2(1+4x+2x^2)y = 8x^2.$$

1) and that $x^2 y = z$, show that

$$\frac{d^2 z}{dx^2} + 4 \frac{dz}{dx} + 4z = 8x^2.$$

[4]

2) Find the **general solution** for y in terms of x .

[8]

Describe the behaviour of y as $x \rightarrow \infty$.

[2]

$$(Ans: y = \frac{A}{x^2} e^{-2x} + \frac{B}{x} e^{-2x} + 2 - \frac{4}{x} + \frac{3}{x^2}; y \rightarrow 2)$$

Solution

$$1) \quad z = x^2 y$$

$$\frac{dz}{dx} = x^2 \frac{dy}{dx} + 2xy$$

$$\begin{aligned} \frac{d^2 z}{dx^2} &= x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + 2x \frac{dy}{dx} + 2y \\ &= x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y \end{aligned}$$

Given,

$$x^2 \frac{d^2 y}{dx^2} + 4x(1+x) \frac{dy}{dx} + 2(1+4x+2x^2)y = 8x^2$$

$$x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 4x^2 \frac{dy}{dx} + 2y + 8xy + 4x^2 y = 8x^2$$

$$\left[x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y \right] + 4 \left[x^2 \frac{dy}{dx} + 2xy \right] + 4(x^2 y) = 8x^2$$

$$\frac{d^2 z}{dx^2} + 4 \frac{dz}{dx} + 4z = 8x^2 \quad [\text{shown}]$$

2) Auxiliary Eqn

$$\lambda^2 + 4\lambda + 4 = 0$$

$$(\lambda + 2)^2 = 0$$

$$\lambda = -2 \quad [\text{repeated}]$$

Complementary func.

$$z_c = A e^{-2x} + B x e^{-2x}$$

$$= (A + Bx) e^{-2x}$$

Particular Integral

$$z_p = Px^2 + Qx + R$$

$$z_p' = 2Px + Q$$

$$z_p'' = 2P$$

$$\therefore z_p'' + 4z_p' + 4z_p = 8x^2$$

$$2P + [8Px + 4Q] + 4Px^2 + 4Qx + R = 8x^2$$

$$(4P)x^2 + (8P + 4Q)x + (2P + 4Q + R) = 8x^2 + 0x + 0$$

By comparison

$$4P = 8 \Rightarrow P = 2$$

$$8P + 4Q = 0 \Rightarrow Q = -4$$

$$2P + 4Q + R = 0 \Rightarrow R = 3$$

 $\therefore$ Particular Integral is

$$z_p = 2x^2 - 4x + 3$$

General Soln

$$z = z_c + z_p$$

$$z = (A + Bx) e^{-2x} + 2x^2 - 4x + 3$$

$$y = \frac{1}{x^2} [(A + Bx) e^{-2x} + 2x^2 - 4x + 3] \neq$$

$$\text{As } x \rightarrow \infty, \frac{4}{x} \rightarrow 0, \frac{3}{x^2} \rightarrow 0, e^{-2x} \rightarrow 0$$

$$\therefore y = \frac{1}{x^2} [0 + 2x^2 - 4x + 3]$$

$$= \left[2 - \frac{4}{x} + \frac{3}{x^2} \right]$$

$$y = [2 - 0 - 0]$$

$$= 2 \neq$$

Example 13 (CIE 2019w11/11E)

It is given that $w = \cos y$ and

$$\tan y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 2 \tan y \frac{dy}{dx} = 1 + e^{-2x} \sec y.$$

(a) Show that

$$\frac{d^2 w}{dx^2} + 2 \frac{dw}{dx} + w = -e^{-2x}.$$

[4]

(b) Find the **particular solution** for y in terms of x , given that when $x = 0$, $y = \frac{1}{3}\pi$ and $\frac{dy}{dx} = \frac{1}{\sqrt{3}}$.

[10]

$$(\text{Ans: (b) } y = \cos^{-1}\left(\left(\frac{3}{2} - x\right)e^{-x} - e^{-2x}\right))$$

Solution

Given,

$$a) \quad w = \cos y$$

$$\frac{dw}{dx} = -\sin y \frac{dy}{dx}$$

$$\frac{d^2 w}{dx^2} = -\sin y \frac{d^2 y}{dx^2} - \cos y \left(\frac{dy}{dx}\right)^2$$

$$\tan y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 2 \tan y \frac{dy}{dx} = 1 + e^{-2x} \sec y$$

$$-\sin y \frac{d^2 y}{dx^2} - \cos y \left(\frac{dy}{dx}\right)^2 - 2 \left[\sin y \frac{dy}{dx}\right] = -\cos y - e^{-2x}$$

$$\left[-\sin y \frac{d^2 y}{dx^2} - \cos y \left(\frac{dy}{dx}\right)^2\right] - 2 \left[\sin y \frac{dy}{dx}\right] + \cos y = -e^{-2x}$$

$$\frac{d^2 w}{dx^2} - 2 \frac{dw}{dx} + w = -e^{-2x}$$

2) Auxiliary Eqn

$$\lambda^2 + 2\lambda + 1 = 0$$

$$(\lambda + 1)^2 = 0$$

$$\lambda = -1 \text{ [repeated]}$$

Complementary func.

$$z_c = Ae^{-x} + Bxe^{-x}$$

$$= (A + Bx)e^{-x}$$

Particular Integral

$$w_p = Pe^{-2x}$$

$$w_p' = -2Pe^{-2x}$$

$$w_p'' = 4Pe^{-2x}$$

$$\therefore w_p'' + 2w_p' + w_p = -e^{-2x}$$

$$4Pe^{-2x} - 4Pe^{-2x} + Pe^{-2x} = -e^{-2x}$$

$$[P]e^{-2x} = [-1]e^{-2x}$$

By comparison

$$P = -1$$

$\therefore$ Particular Integral is

$$w_p = -e^{-2x}$$

General Soln

$$w = w_c + w_p$$

$$w = (A + Bx)e^{-x} - e^{-2x}$$

$$\frac{dw}{dx} = -(A + Bx)e^{-x} + Be^{-x} + 2e^{-2x}$$

$$\text{When } x = 0, y = \frac{\pi}{3}, \frac{dy}{dx} = \frac{1}{\sqrt{3}}$$

$$w = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\therefore \frac{1}{2} = -(A + 0)(1) + 0 + 2$$

$$\boxed{A = \frac{3}{2}}$$

$$\frac{dw}{dx} = (-\sin \frac{\pi}{3})\left(\frac{1}{\sqrt{3}}\right) = -\frac{1}{2}$$

$$\therefore -\frac{1}{2} = -\left(\frac{3}{2} + 0\right)(1) + B(1) + 2(1)$$

$$\boxed{B = -1}$$

Particular soln

$$w = \left(\frac{3}{2} - x\right)e^{-x} - e^{-2x}$$

$$y = \cos^{-1}\left[\left(\frac{3}{2} - x\right)e^{-x} - e^{-2x}\right] \quad \#$$

